

DIHEDRAL GROUPS

A PROJECT REPORT

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by

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Abstract

Group Theory is among the most central concepts in Abstract Algebra. It involves the study of algebraic structures called 'groups'. Group Theory provides a framework to formally study symmetry of structures as well as analyze them in an efficient manner.

In this document, we analyze the symmetries of the *regular n -gon* which corresponds to the Dihedral Group, D_{2n} where n is the number of vertices in the *regular n -gon*. We first explore some introductory results in Group Theory and later formally define the Dihedral Group. We view the Dihedral Group as consisting of reflections and rotations and as we will prove these are the two distinct classes of symmetries of a *regular n -gon*. We will also show that every symmetry of a *regular n -gon* is a product of a reflection and a rotation. We then study the structure of the Dihedral Group specifically its semi-direct form. Finally, we show that the automorphism group of the Dihedral Groups is structurally identical to $\mathbb{Z}_n \rtimes \mathbb{Z}_n^*$.

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Preliminaries and Notation

$a|b, a, b \in \mathbb{Z}$

$a \sim b, a, b \in X, X$ is a set

Binary Operation, $*$

$\oplus_n$

$(G, *)$

HK , where H and K are subsets of groups

$H \leq G$

$[G : H]$

$Z(G)$

$a \in G, C_G(a)$

$A \subseteq G, C_G(A)$

$N \trianglelefteq G$

$cl(a), a \in G$

regular n -gon

a is divisible by $b, b = ka, k \in \mathbb{Z}$

$(a, b) \in R \subseteq X \times X$

A map on set X such that $*$: $X \times X \rightarrow X$

Addition modulo n

G is a group under the binary operation $*$

$HK = \{hk \mid h \in H, k \in K\}$

H is a subgroup of G

The index of subgroup $H \leq G$.

$Z(G) = \{a \in G \mid ag = ga \forall g \in G\}$

$C_G(a) = \{g \in G \mid ag = ga\}$

$C_G(A) = \{g \in G \mid ag = ga, \forall a \in A\}$

$gNg^{-1} \leq N \forall g \in G$

Conjugacy class of $a, cl(a) = \{g^{-1}ag \mid g \in G\}$

a polygon with n sides of equal length and

all interior angles of same measure.

Euclid's Division Lemma : For any integer $a \in \mathbb{Z}$, and positive integer $b \in \mathbb{N}$, there exists unique integers $q, r \in \mathbb{Z}$ with $0 \leq r < b$ such that $a = bq + r$.

Well-Ordering Principle: Every non-empty set of positive integers contains a smallest number.

Chapter 1

Introduction and Definitions

1.1 Groups and Subgroups

Definition 1 (Group). *A non-empty set G with a binary operation $*$: $G \times G \rightarrow G$ is a Group if the following conditions are satisfied :*

- Associativity : *For all $a, b, c \in G$. we have $a * (b * c) = (a * b) * c$.*
- Identity : *There exists $e \in G$ such that $e * a = a * e$, for all $a \in G$.*
- Inverse : *For each $a \in G$ there exists $b \in G$ such that $a * b = b * a = e$.*

We will denote the Group G under the binary operation $*$ as $(G, *)$.
When there is no ambiguity regarding the binary operation, we will simply write G .

Some common Groups include : $(\mathbb{Z}, +)$, $(\mathbb{R} \setminus \{0\}, \times)$, $(\mathbb{Z}_n, \oplus_n)$,

$GL(n, \mathbb{R})$ - Group of $n \times n$ invertible matrices over $\mathbb{R}$ under matrix multiplication,

D_{2n} - Group of symmetries of a regular n -gon under composition.

From the associativity law, we can write

$$\underbrace{a * a * \cdots * a}_{n \text{ times}} = a^n$$

Definition 2 (order of elements). *Given a Group $(G, *)$, for any element $a \in G$. The order of a in G is defined as the least positive integer n such that*

$$a^n = e$$

we denote the order of a in G as $|a|$.

Lemma 1. Let $a \in G$, $|a| = n$ and $l, m \in \mathbb{N}$.

- If $a^l = e$, then $n \mid l$.
- $|a^m| = \frac{n}{\gcd(m, n)}$

Proof. (i) Assume $n \nmid l$ then by Euclid's Division Lemma, $l = nq + r$ where $q, r \in \mathbb{Z}$ and $0 < r < n$.

$$e = a^l = a^{nq+r} = a^{nq} * a^r = e * a^r = a^r$$

This contradicts the minimality of the order of a . Thus, $n \mid l$.

(ii) Let $d = \gcd(m, n)$ and $|a^m| = t$. We have $a^{mt} = e \Rightarrow n \mid mt \Rightarrow \frac{n}{d} \mid \frac{m}{d} t$.

As $\gcd(n, m) = d$ we have $\gcd(\frac{n}{d}, \frac{m}{d}) = 1$. Therefore,

$$\frac{n}{d} \mid t$$

Now, $(a^m)^{\frac{n}{d}} = (a^n)^{\frac{m}{d}} = e$. Thus,

$$t \mid \frac{n}{d}$$

Hence, $|a^m| = \frac{n}{\gcd(m, n)}$. □

Definition 3 (Commuting Elements and Abelian Groups). If $a * b = b * a$ for some $a, b \in (G, *)$. We say that a and b commute. If every pair of elements in $(G, *)$ commute with each other then $(G, *)$ is an Abelian Group.

Definition 4 (Subgroups). Given a Group $(G, *)$. A subset H of G is a subgroup of G if H itself is a Group under $*$. We will denote this by $H \leq G$

Note : H must be a group under the same operation as G .

- The subgroup $H = \{e\}$ is the trivial subgroup.
- H is a 'proper subgroup' if H is a proper subset. In this case, we will write it as $H \subsetneq G$.

Theorem 1. Given a collection of subgroups $\{H_1, H_2, \dots, H_n\}$, $n \in \mathbb{N}$ and each $H_i \leq G$ for $i \in \{1, 2, \dots, n\}$ then

$$H = \bigcap_{i \in I} H_i \leq G$$

Proof. We know $e \in H_i$ for all $i \in I$ as $H_i \leq G$. Therefore the identity exists and we have $e \in H \Rightarrow H \neq \phi$.

Let $a, b \in H \Rightarrow a, b \in H_i$ for all $i \in I$. We get,

$$a * b \in H_i, \forall i \in I$$

Therefore, $a * b \in H$.

Let $a \in H$ then $a \in H_i$ for all $i \in I$. As $H_i \leq G$,

$$a^{-1} \in H_i, \forall i \in I$$

We get $a^{-1} \in H$. Therefore, $H \leq G$. □

Example 1. Some subgroups of groups are listed below :

- $2\mathbb{Z}$ is a subgroup of $(\mathbb{Z}, +)$
- The set of all rotation in D_{2n} form a subgroup of D_{2n}
- The Special Linear Group, $SL(n, \mathbb{R})$ (set of matrices with determinant of 1 or -1) is a subgroup of $GL(n, \mathbb{R})$.

1.2 Special Subgroups

For the sake of convenience, we will write any group $(G, *)$ simply as G and the product of elements as ab instead of $a * b$ when there is no ambiguity regarding the group operation.

Definition 5 (Center of a Group, $Z(G)$). The center of a group G denoted by $Z(G)$ is the set

$$Z(G) = \{ g \in G \mid gx = xg, \forall x \in G \}$$

We now prove that $Z(G) \leq G$.

Proof. To check that $Z(G)$ is a subgroup of G . It immediately follows that $Z(G) \subseteq G$ and associativity follows from Group operation .

Now

$$e * x = x * e \text{ for all } x \in G$$

Thus, $e \in Z(G)$. Now let $a \in Z(G)$ for all $x \in G$. We get,

$$ax = xa \Rightarrow a^{-1}(ax)a^{-1} = a^{-1}(xa)a^{-1} \Rightarrow xa^{-1} = a^{-1}x$$

The above results is true for all x in G . Therefore, $a^{-1} \in Z(G)$.

Let $a, b \in Z(G)$. Then for all x in G .
 $ax = xa$ and $bx = xb$. Now ,

$$(ab)x = a(bx) = a(xb) = (ax)b = (xa)b = x(ab)$$

Hence, we get $ab \in Z(G)$ for all $a, b \in Z(G)$.

As $Z(G)$ is a group and also a subset of G . We get the following result, $Z(G) \leq G$. □

Some other subgroups include :

Definition 6 (Centralizers, $C_G(A)$). *Given a group G and a non-empty subset A . The centralizer of A in G , $C_G(A)$ is defined as*

$$C_G(A) = \{g \in G \mid ga = ag, \forall a \in A\}$$

It is the set of all elements in G that commute with A . If A is a singleton set the definition still holds.

We note that if $A = G$ then $C_G(G) = Z(G)$.

Definition 7 (Normalizers, $N_G(A)$). *For any subset A of G , we define $gAg^{-1} = \{gag^{-1} \mid a \in A\}$. The Normalizer of A in G is*

$$N_G(A) = \{g \in G \mid gAg^{-1} = A\}$$

From the above definition, we can define a special subgroup called the Normal subgroup.

Definition 8 (Normal Subgroups). *A subgroup N of G is a Normal Subgroup of G is*

$$N_G(N) = G$$

$$gNg^{-1} = N \quad \forall g \in G$$

It is easy to check that the subgroups defined above are indeed subgroups of G .

1.3 Lagrange's Theorem and Cosets

Definition 9 (Cosets). *If $H \leq G$. For any $g \in G$,*

$$gH = \{gh \mid h \in H\}$$

The set gH is a left coset of H , the right coset Hg is similarly defined.

Note: *The cosets may not necessarily be a Group.*

The reasoning followed for left cosets may also be applied to right cosets. Hence, the results derived in this section apply to both left cosets and right cosets. Throughout this section when referring to cosets we consider the left cosets unless specified.

Theorem 2 (Cosets Partition the Group). *Let $H \leq G$. The set of left cosets of H in G partition G .*

Proof. We note that as H is a subgroup, $e \in H$. We get that $\forall g \in G, g \in gH$.

$$G = \bigcup_{g \in G} gH$$

Claim : If $a, b \in G$, then $aH = bH$ or $aH \cap bH = \phi$.

If the cosets are disjoint we are done, so assume that $x \in aH \cap bH$. Then for some $h_1, h_2 \in H$

$$x = ah_1 = bh_2$$

$$a = bh_2(h_1)^{-1} = bh' \in bH, \quad h' = h_2(h_1)^{-1} \in H$$

Then for any $ah \in aH$,

$$ah = (bh')h = b(hh') \in bH$$

Therefore, $aH \subseteq bH$, a similar calculation by multiplying h_2^{-1} yields that $bH \subseteq aH$. We get $aH = bH$.

From the above derivation, we see that the left cosets of H do partition G .

Remark: We note that equality of any two cosets, uH and vH is true if and only if u and v are representatives of the same coset. $\square$

As the cosets partition Group G , we can define the index of a subgroup.

Definition 10 (Index of a subgroup). *Let $H \leq G$. then the index of H in G is defined as the number of cosets of H in G . We shall denote the index by $[G : H]$.*

Theorem 3 (Lagrange's Theorem). *Given a Finite Group G and subgroup H of G . The order of subgroup H divides the order of G . In fact*

$$|G| = [G : H] \times |H|$$

Proof. Given any $g \in G$, consider the map $f : H \rightarrow gH$ where $f(h) = gh$, $h \in H$. We see that f is surjective by definition, if

$$f(h_1) = f(h_2) \Rightarrow gh_1 = gh_2$$

$$g^{-1}(gh_1) = g^{-1}(gh_2) \Rightarrow h_1 = h_2$$

Therefore f is a bijection. We get $|H| = |gH|$, any coset of H in G has the same cardinality.

By Theorem (2), $|G| = [G : H] \times |H|$. $\square$

Definition 11 (Cyclic Group). We define the cyclic group generated by a as:

$$G = \langle a \rangle = \{a^m \mid m \in \mathbb{Z}\}$$

- Note that $|G| = n$ if and only if $|a| = n$.

Corollary 1. Given a Group G , let $a \in G$. Then $a^{|G|} = e$.

Proof. The order of subgroup $H = \langle a \rangle$ divides $|G|$ by Theorem (3). But $|H| = |a|$,

$$|G| = |a| \times k, \quad k \in \mathbb{N}$$

Therefore, $a^{|G|} = e$.

□

1.4 Quotient Groups

Given a Normal Subgroup Group $N \trianglelefteq G$, we can define a Group consisting of the cosets of N under the operation $\circ$,

$$(uN) \circ (vN) = (uv)N \tag{1.1}$$

Where $u, v \in G$.

Lemma 2. Given a subgroup $N \leq G$. The coset operation defined by,

$$(uN) \circ (vN) = (uv)N$$

where $u, v \in G$ is well defined if and only if N is Normal in G .

Proof. ($\Rightarrow$) Assume that the coset operation is well-defined. Then for any $u, u_1 \in uN$ and $v, v_1 \in vN$. We get,

$$uvN = u_1v_1N$$

Choosing arbitrary $g \in G$ and $n \in N$. Taking $u = e, u_1 = n$ and $v = v_1 = g^{-1}$. As the coset operation is well-defined we have,

$$g^{-1}N = ng^{-1}N$$

Now, $e \in N$, so $ng^{-1} = g^{-1}n' \in g^{-1}N$.

We get $gng^{-1} = n' \in N$, for every $g \in G$ and $n \in N$.

Thus N is a normal subgroup of G .

($\Leftarrow$) Let $N \trianglelefteq G$. Take $u, u_1 \in uN$, and $v, v_1 \in vN$. For some $n_1, n_2 \in N$,

$$u_1 = un_1, \quad v_1 = vn_2$$

Then,

$$u_1v_1 = (un_1)(vn_2) = u(vv^{-1})n_1vn_2 = (uv)(v^{-1}n_1v)n_2$$

As $N \trianglelefteq G$, $v^{-1}n_1v = n' \in N$,

$$u_1v_1 = (uv)(n'n_2) \in (uv)N$$

By the remark in Theorem (2), u_1v_1 are the representatives of the coset uvN .
Therefore for $uN = u_1N$ and $vN = v_1N$,

$$(uN) \circ (vN) = (uv)N$$

$$(u_1N) \circ (v_1N) = (u_1v_1)N$$

$$(uvN) = (u_1v_1)N$$

The coset operation is well-defined. □

Theorem 4 (Quotient Group). *Given a Normal Subgroup Group $N \trianglelefteq G$.*

$$G/N = \{gN \mid g \in G\}$$

is a group under the coset operation on the cosets of N defined as $(uN) \circ (vN) = (uv)N$ where $u, v \in G$.

Proof. Closure and associativity follow from the group operation of G .

existence of identity

As $e \in G$. For any coset $gN \in G/N$,

$$(gN) \circ (eN) = (g * e)N = gN$$

$$(eN) \circ (gN) = (e * g)N = gN$$

Therefore the identity of G/N is $N = eN \in G/N$.

Existence of inverse

Let $gN \in G/N$, $g \in G \Rightarrow g^{-1} \in G$,

$$(gN) \circ (g^{-1}N) = (gg^{-1})N = N = (g^{-1}g)N = (g^{-1}N) \circ (g)N$$

□

1.5 Homomorphisms And Isomorphisms

Definition 12 (Homomorphisms). *A map $\phi : (G_1, *_1) \rightarrow (G_2, *_2)$ is a 'Homomorphism' if for all $a_1, b_1 \in G_1$.*

$$\phi(a_1 *_1 b_1) = \phi(a_1) *_2 \phi(b_1)$$

The Homomorphism $\Phi : G \rightarrow H$. $\Phi(g) = e_H$, $\forall g \in G$ where e_H is the identity in H is the 'trivial' homomorphism.

Given a homomorphism $\phi : (G, *) \rightarrow (H, \cdot)$ We note the following properties:

- $\phi(e_G) = e_H$. Take any $g \in G$

$$\begin{aligned}\phi(g) &= \phi(g * e_G) = \phi(g) \cdot \phi(e_G) \\ (\phi(g))^{-1} \cdot \phi(g) &= (\phi(g))^{-1} \cdot \phi(g) \cdot \phi(e_G) \\ \phi(e_G) &= e_H\end{aligned}$$

Definition 13 (Isomorphism and Isomorphic Groups). *The map $\phi : (G_1, *_1) \rightarrow (G_2, *_2)$ is an 'Isomorphism'. If for all $a_1, b_1 \in G_1$,*

$$\phi(a_1 *_1 b_1) = \phi(a_1) *_2 \phi(b_1)$$

And the map ϕ is a bijection.

If we can define such a map ϕ between two Groups G_1 and G_2 , then the two Groups are isomorphic.

We will denote two groups as isomorphic by $G_1 \cong G_2$.

Theorem 5. *Isomorphism is an equivalence relation given by,*

$$G_1 \sim G_2 \Leftrightarrow G_1 \cong G_2$$

Proof. Clear $G_1 \cong G_1$ (reflexive) as the identity map(id) is bijective and

$$id(x * y) = x * y = id(x) * id(y)$$

If $G_1 \sim G_2$, then there exists a bijection $f : (G_1, *) \rightarrow (G_2, \cdot)$ such that,

$$f(a * b) = f(a) \cdot f(b), \forall a, b \in G_1$$

As f is a bijection the inverse $f^{-1} : (G_2, \cdot) \rightarrow (G_1, *)$ is also a bijection where For any $x, y \in G_2$ taking $f(m) = x, f(n) = y$ ($m, n \in G_1$ exists as f is a bijection).

$$f^{-1}(x \cdot y) = f^{-1}(f(m) \cdot f(n)) = f^{-1}(f(m * n)) = m * n = f^{-1}(x) * f^{-1}(y)$$

Thus , $G_2 \sim G_1$ (symmetric).

Now if $G_1 \sim G_2$ and $G_2 \sim G_3$. There exists isomorphisms $f : (G_1, *) \rightarrow (G_2, \cdot)$ and $g : (G_2, \cdot) \rightarrow (G_3, +)$, The map $g \circ f : (G_1, *) \rightarrow (G_3, +)$ is a bijection as it is the composition of two bijective maps.

Now for any $a, b \in G_1$,

$$\begin{aligned}(g \circ f)(a * b) &= g(f(a * b)) = g(f(a) \cdot f(b)) = g(f(a)) + g(f(b)) \\ (g \circ f)(a * b) &= (g \circ f)(a) + (g \circ f)(b)\end{aligned}$$

Thus $(g \circ f)$ is an isomorphism and we get $G_1 \sim G_3$.

Remark: If $G_1 = G_2 = G_3 = G$ then we have an isomorphism from G to itself.

Therefore, the isomorphism relation is an equivalence relation.

□

Given below are some examples of homomorphisms and isomorphic groups.

Example 1 : Consider $f : (\mathbb{R}, +) \rightarrow (\mathbb{R}^+, *)$,

$$f(x) = e^x$$

. The map is bijective as the natural logarithm is well defined for any non-zero positive real number. Also for any $x, y \in \mathbb{R}$,

$$e^{x+y} = e^x * e^y$$

Thus, $(\mathbb{R}, +) \cong (\mathbb{R}^+, *)$.

Example 2 : Let $G = \langle a \rangle$ be a cyclic group.

Assume that order of G is n . Consider the additive (under mod n) group of $\mathbb{Z}_n = \{i \in \mathbb{Z} \mid 0 \leq i \leq n-1\}$.

The map $\phi : G \rightarrow \mathbb{Z}_n$, $\phi(a^m) = m(\text{mod } n)$. The map ϕ is a bijection.

And for any $a^x, a^y \in G$.

$$\phi(a^x * a^y) = (x + y)(\text{mod } n) = (x(\text{mod } n)) + (y(\text{mod } n)) = \phi(x) + \phi(y)$$

Therefore, given any order n cyclic group G .

$$G \cong \mathbb{Z}_n$$

1.6 Automorphism Group

Definition 14 (Automorphism). *An isomorphism $\phi : G \rightarrow G$ from a group G to itself is an 'automorphism of G '.*

In fact, if we consider the set of all automorphisms of G , under the binary operation of function composition we get that the set of automorphisms forms a group.

Theorem 6. *The set of automorphisms of a group G under composition forms a group. We denote this group by $\text{Aut}(G)$.*

Proof. We note that associativity follows from the operation and closure follows from the remark in Theorem (5). We need to only prove the existence of identity and inverse.

The identity map $\text{id} : G \rightarrow G$ where $\text{id}(g) = g$, $\forall g \in G$. It is easy to check that id is indeed an automorphism. We also note, for any $\phi \in \text{Aut}(G)$ and $g \in G$ we have,

$$\phi \circ \text{id}(x) = \text{id} \circ \phi(x) = \phi(x)$$

Thus, the identity in $\text{Aut}(G)$ is the identity map.

We now consider an element $f \in \text{Aut}(G)$, by the reasoning used in (5) we note that the inverse function of f , $f^{-1} \in \text{Aut}(G)$. By the definition of a function inverse, we have,

$$f^{-1} \circ f = f \circ f^{-1} = \text{id}$$

We conclude that $\text{Aut}(G)$ is a group under composition. □

Chapter 2

The Dihedral Group D_{2n}

In order to define the Dihedral Group, we must define a symmetry of a *regular n -gon*. We assume that $n \geq 3$

Definition 15 (Symmetry of a regular n -gon). A '*Symmetry*' is a bijection $\sigma : V \rightarrow V$, where V is the set of vertices of the regular n -gon such that σ preserves the adjacency of the vertices.

If we define an adjacency relation that is symmetric, $\sim$. We denote a adjacent to b for any two vertices as $a \sim b \equiv b \sim a$.

We say that σ preserves adjacency if,

$$a \sim b \Rightarrow \sigma(a) \sim \sigma(b)$$

Definition 16 (Dihedral Group D_{2n}). The Dihedral Group D_{2n} is defined as the Group consisting of the symmetries of a regular n -gon under composition.

Given a *regular n -gon*, we will adopt the following convention to index the vertices throughout the chapter.

- We will choose a vertex v , we will denote v by 0.
- Now, proceeding in a clockwise sense we will number the subsequent vertices by $1, 2, 3, \dots, n-1$.
- We note that by this convention, the vertex indexed 0 is adjacent to $n-1$ and 1.

$$0 \sim n-1 \quad \text{and} \quad 0 \sim 1 \tag{2.1}$$

- Thus, our indexed vertex set is $V = \{0, 1, 2, \dots, n-1\}$.

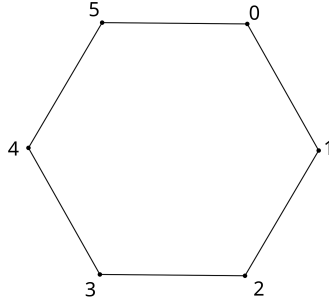


Figure 2.1: Indexing the vertices of a *regular 6-gon* (hexagon)

2.1 Elements in D_{2n}

In this section we consider rotations and reflections of a *regular n -gon*.

We will denote the set of indexed vertices of a *regular n -gon* by $V = \{0, 1, 2, 3, \dots, n-1\}$.

2.1.1 Rotations in D_{2n}

We consider the following map where $k \in \{0, 1, 2, \dots, n-2, n-1\}$,

$$r_k : V \rightarrow V$$

For any $i \in V$,

$$r_k(i) = (i + k)(\text{mod } n)$$

The above map is well-defined for any integer k and we will later prove that the distinct rotations are produced for $k \in \{0, 1, 2, \dots, n-2, n-1\}$. We note that $e = r_0$ for $k = 0$ is the identity as it fixes all the points.

Claim 1. $r_k : V \rightarrow V$ is a bijection for all $k \in \mathbb{Z}$

Proof. The reasoning used applies for arbitrary $k \in \mathbb{Z}$.

We first prove injectivity,

Assume that $r_k(x) = r_k(y)$, $x, y \in V$.

$$(x + k)(\text{mod } n) = (y + k)(\text{mod } n)$$

$$(x + k) \equiv (y + k)(\text{mod } n)$$

$$x \equiv y(\text{mod } n)$$

We know that $x, y \in V$. Thus, $0 \leq x, y < n$ and we get,

$$x = y$$

Hence, r_k is injective. Now, $|V| = n$ and the map r_k is injective.

Therefore, it follows that r_k is a bijection for $k \in \mathbb{Z}$. □

Claim 2. *There are only n distinct bijections r_k where $k \in \{0, 1, 2, \dots, n-1\}$.*

Proof. Assume that $k' \in \mathbb{Z}$. By Euclid's Division Lemma,

$$k' = nq + k, \text{ where } q, k \in \mathbb{Z}, 0 \leq k < n$$

For any vertex $i \in V$,

$$\begin{aligned} r'_k(i) &= (i + k')(\text{mod } n) \\ r'_k(i) &= (i + nq + k)(\text{mod } n) \\ r'_k(i) &= (i + k)(\text{mod } n) \\ r'_k(i) &= r_k(i) \end{aligned}$$

From the above calculations for any $k' \in \mathbb{Z}$, r'_k is equivalent to some r_k where $k \in \{0, 1, 2, \dots, n-1\}$.

To prove that each of the r_k , $k \in \{0, 1, 2, \dots, n-1\}$ are distinct, Assume that $r_{k_1} = r_{k_2}$, $k_1, k_2 \in \{0, 1, 2, \dots, n-1\}$. For any $i \in V$,

$$\begin{aligned} r_{k_1}(i) &= r_{k_2}(i) \\ (i + k_1)(\text{mod } n) &= (i + k_2)(\text{mod } n) \\ (i + k_1) &\equiv (i + k_2)(\text{mod } n) \end{aligned}$$

Therefore,

$$k_1 \equiv k_2(\text{mod } n)$$

Equality follows as $k_1, k_2 \in \{0, 1, 2, \dots, n-1\}$. □

Theorem 7. *For $k \in \{0, 1, 2, \dots, n-1\}$, r_k is a symmetry of a regular n -gon.*

Proof. By *Claim 1* we have that for any k , r_k is a bijection. We must prove that r_k preserves the adjacency of the vertices.

We note that by our indexing convention, any two vertices $i, j \in V$.

$$i \sim j \Leftrightarrow j \equiv i \pm 1 \pmod{n}$$

In fact given a vertex i there are only two adjacent vertices in a *regular n -gon*, these are: $(i + 1)(\text{mod } n)$ and $(i - 1)(\text{mod } n)$.

Consider i and $(i + 1)(\text{mod } n)$, we note that $i \sim (i + 1)(\text{mod } n)$.

$$\begin{aligned} r_k((i + 1)(\text{mod } n)) &= (i + 1 + k)(\text{mod } n) \\ r_k((i + 1)(\text{mod } n)) &= ((i + k)(\text{mod } n) + 1)(\text{mod } n) \\ r_k((i + 1)(\text{mod } n)) &= r_k(i) + 1(\text{mod } n) \end{aligned}$$

Therefore, $r_k(i) \sim r_k((i+1)(\text{mod } n))$.

For the adjacent vertices i and $(i-1)(\text{mod } n)$.

$$r_k((i-1)(\text{mod } n)) = ((i-1) + k)(\text{mod } n) = ((i+k)(\text{mod } n) - 1)(\text{mod } n)$$

$$r_k((i-1)(\text{mod } n)) = (r_k(i) - 1)(\text{mod } n)$$

From the above calculations, r_k preserves adjacency of vertices for any $k \in \{0, 1, 2, \dots, n-1\}$.

Each of the n distinct r_k 's are a valid symmetry of a regular n -gon. □

Theorem 8. *Given any vertex $i \in V$ and $k \in \{0, 1, 2, \dots, n-1\}$.
 $\circ$ is function composition.*

$$r_k = \underbrace{r_1 \circ \dots \circ r_1}_{k \text{ times}} = r_1^k$$

Proof. We shall proceed by induction,

Base Case: When $k = 1$, $r_1 = r_1$ is trivially true.

When, $k = 2$, we have for any $i \in V$,

$$r_1 \circ r_1(i) = r_1((i+1)(\text{mod } n)) = (((i+1)(\text{mod } n)) + 1)(\text{mod } n) = (i+2)(\text{mod } n)$$

Hence, $r_2 = r_1 \circ r_1 = r_1^2$

Inductive Hypothesis: Assume that for some $k \in \mathbb{N}$,

$$r_k = \underbrace{r_1 \circ \dots \circ r_1}_{k \text{ times}} = r_1^k$$

Inductive Step: Now, for $k+1$ and arbitrary $i \in V$,

$$r_{k+1}(i) = (r_1(r_1^k))(i) = (r_1(r_k))(i)$$

$$r_{k+1}(i) = r_1((i+k)(\text{mod } n))$$

$$r_{k+1}(i) = ((i+k)(\text{mod } n) + 1)(\text{mod } n)$$

$$r_{k+1}(i) = (i + (k+1))(\text{mod } n) = r_1^{k+1}$$

By the Principle of Mathematical Induction, $\forall k \in \mathbb{N}$,

$$r_k = r_1^k$$

We see that each r_k can be expressed as successive compositions of r_1 .

It can be easily shown that $\langle r_1 \rangle$ is a cyclic group of order n , using the result derived in Claim (2) and Group Axioms. □

Geometrically visualizing the effect of r_k on a *regular n -gon*, r_k has the effect of a counter-clock-wise rotation by $\frac{2\pi k}{n}$. Since each $r_k = r_1^k$, we will denote r_1 by r . We shall call the map r a '*rotation*' of the *regular n -gon*.

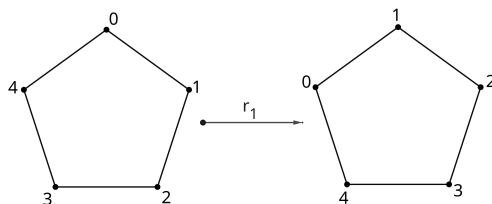


Figure 2.2: Effect of r_1 on a regular 5-gon (pentagon)

We have the following relation as a result of the theorems and claims proved in this section,

$$\langle r \rangle \leq D_{2n}$$

We have $|r| = n$, this fact follows from Claim (2) and Theorem (8).

Note: For any rotation $r_k = r^k$, the inverse of this rotation is,

$$(r^k)^{-1} = r^{(n-k)} = r^{-k}$$

We will write the inverse of r^k as r^{-k} instead of $r^{(n-k)}$.

2.1.2 Reflections in D_{2n}

We now consider another type of map we can define over a *regular n -gon*,

$$s : V \rightarrow V$$

As we progress, we will see that the map s corresponds geometrically to reflecting a *regular n -gon* about an axis passing through vertex 0. Thus, we will refer to s as a '*reflection*'.

Our analysis of this map will be made easier if we consider the cases when:

- n is odd.
- n is even.

Reflections in D_{2n} for odd n

For any vertex $i \in V$,

$$s(i) = (n - i)(\text{mod } n)$$

Claim 3. For any vertex $i \in V$, $s(i) = (n - i)(\text{mod } n)$ is a symmetry.

This result is valid for both odd n and even n .

Proof. In order to prove injectivity, assume that $s(i) = s(j)$ for two vertices $i, j \in V$.

$$s(i) = s(j) \Rightarrow (n - i)(\text{mod } n) = (n - j)(\text{mod } n)$$

$$i \equiv j(\text{mod } n)$$

But as $i, j \in V = \{0, 1, 2, \dots, n - 1\}$. We get,

$$i = j$$

Now, s maps the vertex set V to itself. By the above calculation, s is injective. The bijectivity of the map $s : V \rightarrow V$ follows.

Now consider the pair of adjacent vertices i and $(i + 1)(\text{mod } n)$,

$$s((i + 1)(\text{mod } n)) = (n - ((i + 1)(\text{mod } n)))(\text{mod } n)$$

$$s((i + 1)(\text{mod } n)) = ((n - i)(\text{mod } n) - 1)(\text{mod } n)$$

$$s((i + 1)(\text{mod } n)) = (s(i) - 1)(\text{mod } n)$$

We get $s(i) \sim s((i + 1)(\text{mod } n))$.

A similar reasoning yields the same result for the case $i \sim (i - 1)(\text{mod } n)$.

Therefore, s is a valid symmetry.

□

We consider the following modified map that is also a type of 'reflection',

$$s_k : V \rightarrow V$$

$$s_k = r^k \circ s \circ r^{-k} \tag{2.2}$$

It is immediately obvious that s_k is a bijection. Expanding s_k further, consider $i \in V$,

$$s_k(i) = r^k(s((i + n - k)(\text{mod } n)))$$

$$s_k(i) = r^k((k - i)(\text{mod } n))$$

$$s_k(i) = (2k - i)(\text{mod } n) \tag{2.3}$$

We note that the map s is just a special case of s_k when $k = 0$.

Geometrically, s_k is the reflection about an axis passing through vertex k , it is the reflection that fixes k .

Lemma 3. *Given two symmetries σ and τ . Composition of two symmetries $(\sigma \circ \tau)$ is a symmetry.*

Proof. As the composition of two bijective maps is bijective, we need to only show that $\sigma \circ \tau$ preserves adjacency of vertices.

Take a pair of adjacent vertices $i, j \in V$. As $i \sim j$,

$$\tau(i) \sim \tau(j)$$

We have $\tau(i)$ and $\tau(j)$ as adjacent vertices. Hence,

$$\sigma(\tau(i)) \sim \sigma(\tau(j)) \Rightarrow (\sigma \circ \tau(i)) \sim (\sigma \circ \tau(j))$$

Therefore, $\sigma \circ \tau$ is a symmetry. □

Theorem 9. *Each s_k for $k \in \{0, 1, 2, \dots, n-1\}$ is a distinct symmetry. There are only n such symmetries.*

Proof. By Lemma (3) and (2.2), we get s_k is a symmetry for any $k \in \mathbb{N}$.

We first prove that there are only n such symmetries s_k .

Consider any $k' \in \mathbb{Z}$. By Euclid's Division Lemma, there exists $q, k \in \mathbb{Z}$ where $0 \leq k < n$ such that,

$$k' = nq + k$$

Substituting k' in (2.3), for any $i \in V$,

$$s_{k'}(i) = (2(nq + k) - i)(\text{mod } n)$$

$$s_{k'}(i) = (2nq + 2k - i)(\text{mod } n)$$

$$s_{k'}(i) = s_k(i)$$

Thus, for any $k' \in \mathbb{Z}$, we have $s_{k'} = s_k$ where $k \in \{0, 1, 2, \dots, n-1\}$.

In order to prove each of these symmetries distinct, assume that $s_{k_1} = s_{k_2}$ for some pair of $k_1, k_2 \in \{0, 1, 2, \dots, n-1\}$.

We have, $s_{k_1}(i) = s_{k_2}(i)$, $\forall i \in V$.

$$(2k_1 - i)(\text{mod } n) = (2k_2 - i)(\text{mod } n)$$

$$(2k_1) \equiv (2k_2) \pmod{n}$$

As we are considering an odd n , $\gcd(n, 2) = 1$. Therefore,

$$k_1 \equiv k_2 \pmod{n}$$

Equality follows as $k_1, k_2 \in \{0, 1, 2, \dots, n-1\}$. □

We end our analysis of reflections in D_{2n} for odd n by noting that,

$$\begin{aligned}(r^{2k} \circ s)(i) &= r^{2k}(-i \pmod n) = (2k - i) \pmod n, \quad i \in V \\ (r^{2k} \circ s) &= s_k\end{aligned}$$

As n is odd,

For $k \in \{0, 1, 2, \dots, n-1\}$, $2k$ does generate $\{0, 1, 2, \dots, n-1\}$

We have the following equality,

$$\{r^m \circ s \mid m \in \{0, 1, 2, \dots, n-1\}\} = \{s_k \mid k \in \{0, 1, 2, \dots, n-1\}\}$$

For the sake of notational convenience, when we refer to a reflection s_k we will write it as $r^{2k \pmod n}_s$.

Reflections in D_{2n} for even n

For the case when n is even,

$$s_k = r^k s r^{-k} = r^{2k \pmod n}_s \quad (2.4)$$

Clearly we cannot use the reasoning used in Theorem (9) as n is even.

In fact for an even n , there are only $\frac{n}{2}$ such reflections.

Lemma 4. *For $k \in \{0, 1, 2, \dots, \frac{n}{2}-1\}$, there exists only $\frac{n}{2}$ distinct reflection symmetries of the form $s_k = r^{2k \pmod n}_s$ in D_{2n} for an even n .*

Proof. We note that s_k is a symmetry by Claim (3) and Lemma (3).

We first prove that there are $\frac{n}{2}$ symmetries where $s_k = r^{2k \pmod n}_s$.

For any $k' \in \mathbb{N}$, by Euclid's Division Lemma there exists $q, k \in \mathbb{Z}$ such that $0 \leq k < \frac{n}{2}$ where,

$$k' = q\left(\frac{n}{2}\right) + k$$

We have,

$$\begin{aligned}s'_k &= r^{2k' \pmod n}_s = r^{2(q(\frac{n}{2})+k) \pmod n} \\ s'_k &= r^{(nq+2k) \pmod n}_s = r^{2k \pmod n}_s \\ s'_k &= s_k\end{aligned}$$

Thus, we have at most $\frac{n}{2}$ different symmetries for $0 \leq k < \frac{n}{2}$.

In order to show that each of these symmetries are distinct say $k_1, k_2 \in \{0, 1, 2, \dots, \frac{n}{2}-1\}$ where $s_{k_1} = s_{k_2}$, for $i \in V$,

$$\begin{aligned}s_{k_1}(i) &= s_{k_2}(i) \\ (2k_1 - i) &\equiv (2k_2 - i) \pmod n \\ n &\mid 2(k_1 - k_2) \\ \frac{n}{2} &\mid (k_1 - k_2)\end{aligned}$$

We have $(k_1 - k_2) = c \times \frac{n}{2}$, $c \in \mathbb{Z}$. By assumption that $k_1, k_2 \in \{0, 1, 2, \dots, \frac{n}{2} - 1\}$, we have,

$$0 \leq |k_1 - k_2| < \frac{n}{2}$$

Therefore, $c = 0$ and $k_1 = k_2$, each s_k is distinct. $\square$

Claim 4. For an even n , s_k fixes the vertex k and $k + \frac{n}{2}$ for $k \in \{0, 1, 2, \dots, \frac{n}{2} - 1\}$

Proof. We note that for $k \in \{0, 1, 2, \dots, \frac{n}{2} - 1\}$,

$$\begin{aligned} s_k(k) &= (2k - k)(\text{mod } n) = k(\text{mod } n) = k \\ s_k(k + \frac{n}{2}) &= (2k - (k + \frac{n}{2}))(\text{mod } n) = (k - \frac{n}{2})(\text{mod } n) \\ s_k(k + \frac{n}{2}) &= (k + \frac{n}{2})(\text{mod } n) = k + \frac{n}{2} \end{aligned}$$

The last step follows as $(k + \frac{n}{2} < n)$.

We see that s_k fixes the vertices k and $k + \frac{n}{2}$. $\square$

For an even n we now consider another type of reflection symmetry of the form

$$rs_k = r^{(2k+1)(\text{mod } n)}_s \quad (2.5)$$

where $k \in \{0, 1, 2, \dots, \frac{n}{2} - 1\}$.

Claim 5. For an even n , there are $\frac{n}{2}$ distinct symmetries of the form rs_k , where $k \in \{0, 1, 2, \dots, \frac{n}{2} - 1\}$.

Proof. We note that rs_k is a symmetry by Claim (3) and Lemma (3).

We first prove that there are $\frac{n}{2}$ symmetries where $rs_k = r^{(2k+1)(\text{mod } n)}_s$.

For any $k' \in \mathbb{N}$, by Euclid's Division Lemma there exists $q, k \in \mathbb{Z}$ such that $0 \leq k < \frac{n}{2}$ where,

$$k' = q(\frac{n}{2}) + k$$

We have,

$$\begin{aligned} rs'_k &= r^{(2k'+1)(\text{mod } n)}_s = r^{2(q(\frac{n}{2})+k)+1(\text{mod } n)}_s \\ rs'_k &= r^{(nq+2k+1)(\text{mod } n)}_s = r^{(2k+1)(\text{mod } n)}_s \\ rs'_k &= rs_k \end{aligned}$$

Thus, we have at most $\frac{n}{2}$ different symmetries for $0 \leq k < \frac{n}{2}$.

In order to show that each of these symmetries are distinct say $k_1, k_2 \in \{0, 1, 2, \dots, \frac{n}{2} - 1\}$ where $rs_{k_1} = rs_{k_2}$, for any vertex $i \in V$,

$$rs_{k_1}(i) = rs_{k_2}(i)$$

$$\begin{aligned}
r^{-1}(rs_{k_1})(i) &= r^{-1}(rs_{k_2})(i) \\
(r^{-1}r)s_{k_1}(i) &= (r^{-1}r)s_{k_2}(i) \\
s_{k_1}(i) &= s_{k_2}(i)
\end{aligned}$$

Now for any $k_1, k_2 \in \{0, 1, 2, \dots, n-1\}$. By Lemma (4) if $s_{k_1} = s_{k_2}$ we have $k_1 = k_2$. □

Lemma 5. *For an even n , all the reflection symmetries rs_k where $k \in \{0, 1, \dots, \frac{n}{2} - 1\}$ fixes no vertex.*

Proof. For the sake of a contradiction, let us assume that there exists a $k \in \{0, 1, \dots, \frac{n}{2} - 1\}$ and a vertex $i \in V$ such that rs_k fixes vertex i ,

$$\begin{aligned}
rs_k(i) &= i \\
(2k + 1 - i) &= i \pmod{n}
\end{aligned}$$

We get,

$$nk = 2(k - i) + 1$$

By our assumption that n is even. The above result is a contradiction and no such pair of k and i exist such that rs_k fixes vertex i . □

By lemma (5) we see that for an *even* n . The two reflections (rs_k and s_k) are distinct classes as rs_k fixes no vertex while s_k fixes two vertices. We will see later that for an *even* n these reflections do form two separate equivalence classes.

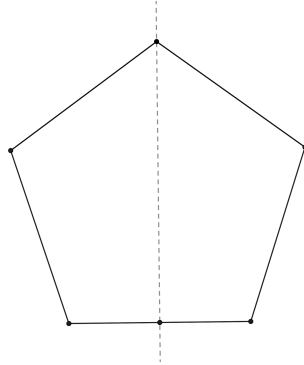


Figure 2.3: The axis of reflection of s_k for *regular n -gon* (odd n) in this case a pentagon

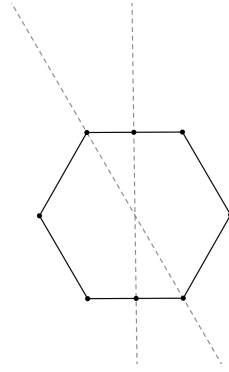


Figure 2.4: The two axis of reflection for *regular n-gon*(even n) in this case a hexagon. The axis through the mid-points of the edges is rs_k while the axis through two opposite vertices is s_k .

2.2 Order of D_{2n}

From our analysis of rotations and reflections in the previous sections. We know that each rotation is distinct and each class of reflections contain distinct reflections. (we are referring to s_k and rs_k).

We have shown that the rotations and reflections are valid symmetries of a *regular n-gon* and are elements of D_{2n} .

Lemma 6. *No reflection is the identity.*

Proof. We consider s_k first ,

$$s_k((k+1)(\text{mod } n)) = (k-1)(\text{mod } n)$$

If s_k were the identity then,

$$k+1 \equiv k-1(\text{mod } n)$$

$$n \mid ((k+1) - (k-1))$$

$$n \mid 2$$

But for a *regular n-gon* we assume that $n \geq 3$. Thus, s_k is not the identity.

We now consider rs_k if it were the identity, a similar analysis yields the same contradiction. $\square$

Claim 6. *Rotations and Reflections are distinct symmetries.*

Proof. Every reflection has an order of 2, for any $i \in V$,

$$(s_k \circ s_k)(i) = s_k(2k - i(\text{mod } n)) = i(\text{mod } n) = i$$

$$(rs_k \circ rs_k)(i) = rs_k((2k+1-i)(\text{mod } n)) = i(\text{mod } n) = i$$

But the only rotation that has order 2 is $r^{\frac{n}{2}}$ for an even n ,

This follows as $|r^m| = \frac{n}{\gcd(n,m)}$, this forces $m = \frac{n}{2}$.

Now $r^k(i) = (i+k)(\text{mod } n)$ does not fix any vertex except when $k \equiv 0(\text{mod } n)$ but $k = 0$ is the identity.

Any s_k always fixes at least one vertex. (for *odd* n any reflection fixes only one vertex)

If some reflection rs_k is equal to another rotation $r_{k'}$,

$$rs_k = r^{k'}$$

For all $i \in V$,

$$(2k+1-i)(\text{mod } n) = (k'+i)(\text{mod } n)$$

$$(2k - k' + 1) \equiv 2i(\text{mod } n)$$

Clearly the above equation holds for all $i \in V$. Choosing the vertices 0 and i ,

$$2 \equiv 0(\text{mod } n)$$

This contradicts our assumption of $n \geq 3$. □

Therefore we have that $\langle r \rangle \cup \{r^m s \mid m \in \mathbb{Z}\} \subseteq D_{2n}$.

By Claim (6) as both the sets are disjoint we have the following inequality,

$$2n \leq |D_{2n}| \tag{2.6}$$

We now prove that the order of D_{2n} is $2n$.

Theorem 10. *The order of D_{2n} is $2n$.*

Proof. Any symmetry of the *regular* n -gon has n choices for sending a vertex to another vertex.

On sending this vertex to its destination there are two choices for its neighbors. Hence, we have fixed an edge and we can describe the entire *regular* n -gon.

By the Product Rule,

$$|D_{2n}| \leq 2n$$

And by equation (2.6), we have the order of D_{2n} as $2n$. □

2.3 Presentation of D_{2n}

In this section we will denote the rotations r_k by r^k and the reflections instead of using the rs_k and s_k notation for reflection we will use the $r^m s$ form. Where if $m = 2k$ it corresponds to s_k and if $m = 2k+1$ it corresponds to rs_k . This follows from directly from the definition of s_k and rs_k respectively.

2.3.1 Relation between rotations and reflections

Let us consider the effect of srs on a regular n -gon. For any $i \in V$,

$$srs(i) = sr(-i \pmod n) = s(-i + 1 \pmod n)$$

$$srs(i) = (i - 1) \pmod n$$

We note that this is precisely $r^{-1}(i)$.

We have the following relation,

$$srs = r^{-1}$$

Lemma 7. *Given an arbitrary sequence of composition of symmetries, $x = x_1 x_2 \dots x_n$, each $x_i \in \{r, s\}$ for $1 \leq i \leq n$. We can re-write x as,*

$$x = r^a s^b$$

where $a \in \mathbb{Z}$ and $b \in \{0, 1\}$.

Proof. We shall proceed by Induction,

Base case: When $n = 1$, $x_1 = rs^0$ or $x_1 = r^0 s$.

For the case where $n=2$ We get,

- $x = rr = r^2 s^0$.
- $x = ss = r^0 s^2$.
- $x = rs$.
- $x = sr = r^{-1} s$.

Inductive Hypothesis: Assume that for $n = k$, we can write $x = r^a s^b$.

Inductive Step: Consider $x' = x'_1 x'_2 \dots x'_k x'_{k+1}$.

$$x' = (x'_1 x'_2 \dots x'_k) x'_{k+1}$$

$$x' = (r^a s^b) x'_{k+1}$$

Now, if $x'_{k+1} = s$. We get $x' = r^a s^{(b+1) \pmod 2}$.

When $x'_{k+1} = r$, $x' = r^a s^b r$.

If $b = 0$, we get $x' = r^{a+1} s^0$.

If $b = 1$, $x' = r^a sr = r^a r^{-1} s = r^{(a-1)} s$.

By the principle of Mathematical Induction, the lemma holds for arbitrary finite product of r and s .

□

We see that any finite sequence of the product of elements can be reduced to $r^a s^b$.

If we consider the group generated by r and s ,

$$\langle r, s \mid srs = r^{-1} \rangle$$

We get the following group,

$$\{e, r, r^2, \dots, r^{n-1}, s, rs, r^2s, \dots, r^{n-1}s\}$$

The above Group contains $2n$ elements and each element is a distinct symmetry of a *regular n -gon*. Using (10) the above group is D_{2n} .

We can write D_{2n} as,

$$D_{2n} = \langle r, s \mid r^n = s^2 = e, srs = r^{-1} \rangle$$

Chapter 3

Centralizers, Conjugacy Classes and Normal Subgroups

3.1 Centralizers of D_{2n}

For any Group G , the Centralizer of an element which we will denote by $C_G(a)$, $a \in G$ is defined as

$$C_G(a) = \{g \in G \mid ag = ga\}$$

We immediately observe that for the identity, $e \in G$.

$$e * g = g * e, \forall g \in G$$

Hence, the centralizer of the identity is the Group itself. For the case where the Group is the Dihedral Group, D_{2n} we get,

$$C_{D_{2n}}(e) = D_{2n}$$

3.1.1 Centralizers of rotations in D_{2n}

Consider the rotation $r^k \in D_{2n}$, where $1 \leq k \leq n-1$,

We note that for any rotation $r^m \in \langle r \rangle$, $m \in \mathbb{Z}$,

$$(r^k)(r^m) = r^{k+m} = r^{m+k} = (r^m)(r^k)$$

Thus, $\langle r \rangle \leq C_{D_{2n}}(r^k)$

Now, we claim that any reflection of the form $r^i s \notin C_{D_{2n}}(r^k)$.

Claim : Any reflection $r^i s \in D_{2n}$ is not in the centralizer of r^k .
Proof : Assume that a reflection, $r^i s \in C_{D_{2n}}(r^k)$, $i \in \mathbb{Z}$.

$$\begin{aligned} r^k(r^i s) &= (r^i s)r^k \\ r^{i+k}s &= sr^{k-i} \Rightarrow sr^{-(i+k)} = sr^{k-i} \end{aligned}$$

$$r^{2k} = e \tag{3.1}$$

From (3.1) and using the fact that $2 \leq 2k \leq 2n - 2$,
The only valid solution for k is $\frac{n}{2}$.

Therefore we get the Centralizers of rotations in D_{2n} as follows :
For any rotation, r^i , $i \neq \frac{n}{2}$: $C_{D_{2n}}(r^k) = \langle r \rangle$

When n is an even integer,
When $k = \frac{n}{2}$: $C_{D_{2n}}(r^{\frac{n}{2}}) = D_{2n}$

3.1.2 Centralizers of reflections in D_{2n}

Consider a reflection $r^i s \in D_{2n}$, $i \in \mathbb{Z}$, $0 \leq i \leq n - 1$.
Following a calculation similar to that of the above subsection.
We get any rotation, $r^k \notin C_{D_{2n}}(r^i s)$, for $k \neq \frac{n}{2}$.

Assume that a reflection $r^j s \in C_{D_{2n}}(r^i s)$, $j \in \mathbb{Z}$, $0 \leq j \leq n - 1$. We get,

$$\begin{aligned} (r^i s)(r^j s) &= (r^j s)(r^i s) \\ r^{i-j} &= r^{j-i} \\ r^{2(j-i)} &= e \end{aligned} \tag{3.2}$$

From (3.2), we get $n|2(j - i)$,

$$2(j - i) \equiv 0 \pmod{n}$$

Case 1 (n is odd) : For an odd n, equation (3.1.2) has the only solution as $i = j$.
The centralizer of $r^i s$ for an odd n,

$$C_{D_{2n}}(r^i s) = \{e, r^i s\}$$

Case 2 (n is even) : For an even n, we know that $|j - i| \leq n - 1 \Rightarrow 2|j - i| \leq 2n - 2$. Hence,

$$2(j - i) = kn, k \in \{-1, 0, 1\}$$

From the above equation we get the solutions

- $k = 0$, which yields $i=j$.
- $k = 1$, which yields $j = i + \frac{n}{2}$.
- $k = -1$ which yields $j = i - \frac{n}{2}$.

We note that $i - \frac{n}{2} \equiv i + \frac{n}{2} \pmod{n}$ because

$$i - \frac{n}{2} + n \equiv i - \frac{n}{2} \pmod{n}$$

The centralizer of $r^i s$ for an even n is

$$C_{D_{2n}}(r^i s) = \{e, r^i s, r^{\frac{n}{2}}, r^{(i+\frac{n}{2})} s\}$$

3.1.3 Center of D_{2n} / Centralizer of D_{2n}

We note that $C_{D_{2n}}(D_{2n}) = Z(D_{2n})$, where $Z(D_{2n})$ is the center of the Group.

From the previous subsections we note that the only element that commutes with the whole Group(D_{2n}) is the identity(e) and when n is even, $r^{\frac{n}{2}}$.

- **n is odd :** $Z(D_{2n}) = \{e\}$
- **n is even :** $Z(D_{2n}) = \{e, r^{\frac{n}{2}}\}$

3.2 Conjugacy Classes of D_{2n}

The conjugacy class of an element($cl(a), a \in G$) in Group, G is defined as

$$cl(a) = \{g^{-1}ag | g \in G\}$$

We will use the fact the Conjugacy classes partition the Group. The equivalence relation is

$$a \sim b \Leftrightarrow \exists g \in G, b = g^{-1}ag$$

We note that the conjugacy class of any element, a in $Z(D_{2n})$ consists of the singleton set containing a .

Therefore, we get

- $cl(e) = \{e\}$
- $cl(r^{\frac{n}{2}}) = \{r^{\frac{n}{2}}\}$, when n is even

3.2.1 Conjugacy Classes of rotations, r^k ($k \neq \frac{n}{2}$)

We will first consider conjugating r^k with a rotation, r^m , $m \in \mathbb{Z}$.

$$(r^m)^{-1}(r^k)(r^m) = r^{-m+k+m} = r^k$$

Now conjugating r^k with a reflection $r^m s$,

$$(r^m s)^{-1}(r^k)(r^m s) = s r^{-m+k+m} s = r^{-k}$$

From the above calculations we get the conjugacy classes of a rotation $r^k \in D_{2n}$, $k \neq \frac{n}{2}$ as

$$cl(r^k) = \{r^k, r^{-k}\}$$

There are exactly $\frac{n-1}{2}$ such classes for an *odd* n ,

And $\frac{n}{2} - 1$ such classes when n is *even*

Now we consider the conjugacy classes of reflections, $r^k s \in D_{2n}$, $k \in \mathbb{Z}$.

3.2.2 Conjugacy Classes of reflections, $r^k s \in D_{2n}$

It will simplify the calculations if we consider 2 cases: an odd n , an even n .

Case 1: n is odd Conjugating $s \in D_{2n}$ with a rotation, $r^m \in D_{2n}$,

$$r^{-m} s r^m = r^{-2m} s$$

We notice that since n is odd, $r^{-2m} s$ will generate the entire set of reflections in D_{2n} .

The conjugacy class of reflection is,

$$cl(s) = \{r^k s | k \in \mathbb{Z}\} \quad (3.3)$$

Case 2: n is even Clearly all the reflections $r^{2m} s, m \in \mathbb{Z}$ form one conjugacy class.

Now consider the conjugate of $rs \in D_{2n}$ with a rotation, $r^m \in D_{2n}$,

$$r^{-m}(rs)r^m = r^{-2m+1} s$$

Therefore,

$$cl(rs) = \{r^{2m+1} s | m \in \mathbb{Z}\}$$

We notice that $cl(s) \cup cl(rs) = \{r^k s | k \in \mathbb{Z}\}$, As Conjugacy classes partition the Group, we note that the union of all the classes does equal D_{2n} .

Listing out all the conjugacy classes of D_{2n} explicitly, we get

Size of the class	n is odd	n is even
1	$\{e\}$	$\{e\}, \{r^{\frac{n}{2}}\}$
$2(1 \leq i \leq n-1, i \neq \frac{n}{2})$	$\{r^i, r^{-i}\}, r^i \in D_{2n}$	$\{r^i, r^{-i}\}, r^i \in D_{2n}$
n	$\{r^k s k \in \mathbb{Z}\}$	
$\frac{n}{2}$		$\{r^{2k} s k \in \mathbb{Z}\}, \{r^{2k+1} s k \in \mathbb{Z}\}$

3.3 Normal Subgroups of D_{2n}

The Normal Subgroup of any Group, G , is the subgroup $N \trianglelefteq G$, where

$$gNg^{-1} \leq N \leq G, \forall g \in G$$

We will first consider the Normal Subgroups on D_{2n} consisting of only rotation and later the subgroups that contain atleast one reflection.

Before proceeding further, we recall the following theorems :

Theorem 11 (Index-2 subgroups). *Let G be a finite Group and H is a subgroup of index-2. Then H is a normal subgroup of G .*

Theorem 12 (Subgroups of Cyclic Groups). *Every Subgroup of a cyclic group is cyclic. In fact for every divisor of a finite cyclic group, there exists a unique cyclic subgroup of that order.*

Theorem 13 (Subgroups of Normal cyclic Groups). *If a cyclic group N is Normal in G , every subgroup of N is Normal in G .*

3.3.1 Normal Subgroups containing only rotations

Now consider the group $\langle r \rangle \leq D_{2n}$, we note that $[D_{2n} : \langle r \rangle] = 2$ by Theorem (11),

$$\langle r \rangle \trianglelefteq D_{2n}$$

By applying Theorem (12) and Theorem (13), we get that every subgroup of $\langle r \rangle$ is $\langle r^d \rangle$ where $d|n$ and $\langle r^d \rangle$ is normal in D_{2n} .

Hence, the only normal subgroups in D_{2n} that contain only rotation are the subgroups of $\langle r \rangle$.

If we denote the number of divisors of n by $\tau(n)$.
There are only $\tau(n)$ such normal subgroups.

3.3.2 Normal Subgroups that contain atleast one reflection

We will now assume that our Normal Subgroup(N) contains atleast one reflection, $r^k s \in D_{2n}$.

To simplify calculations we will consider the cases where n is even and when n is odd.

Case 1: (n is odd) From (3.3), we realize that if any reflection is in N , N must contain all the 'n' reflections. Because N is a subgroup also it must contain the identity. Thus N has $n+1$ elements.

By *Lagrange's Theorem*, $|N| \mid |D_{2n}|$. The only possibility is that N contains $2n$ elements.

Thus, it must be the case that $N = D_{2n}$.

Case 2: (n is even) We recall that :

- $cl(s) = \{r^{2k}s \mid k \in \mathbb{Z}\}$
- $cl(rs) = \{r^{2k+1}s \mid k \in \mathbb{Z}\}$

By a similar reasoning as that in *Case 1*, we note that if N contains two reflections $(r^m s, r^l s)$ such that m is an odd integer and l is an even integer. N must be D_{2n} itself.

Hence, it will suffice to consider that N contains only one of the following conjugacy classes: $cl(s)$ or $cl(rs)$.

Case 2.a : $cl(s) \subseteq N$, Clearly $|N| > \frac{n}{2} \cdot (N \text{ has at least } \frac{n}{2} + 1 \text{ (identity)})$.

$$[D_{2n} : N] = \frac{|D_{2n}|}{|N|} < \frac{2n}{\frac{n}{2}} = 4$$

By *Lagrange's Theorem* and using $|N| > \frac{n}{2}$ we get $|N| = n$.
Now N contains $\frac{n}{2}$ reflections, By assumption that N does not contain $cl(rs)$, these are the only reflections that N contains.

We note that,

$$(r^{2m+2}s)(r^{2m}s) = r^2$$

Thus N contains the subgroup $\langle r^2 \rangle$, $|\langle r^2 \rangle| = \frac{n}{2}$.
 $N = cl(s) \cup \langle r^2 \rangle$. (as $cl(s)$ and $\langle r^2 \rangle$ are disjoint and their cardinality is $\frac{n}{2}$ each).

Claim 7. $N = \langle r^2, s \mid sr^2s = r^{-2} \rangle \cong D_n$

Proof. Let $x \in \langle r^2, s \mid sr^2s = r^{-2} \rangle \Rightarrow x = r^{2m}s^a$, $2m \in \mathbb{Z}$, $a = \{0, 1\}$
We can write x as $r^{2m}s^a$ because

$$(sr^2s)^k = sr^{2k}s = r^{-2k}$$

$$sr^{2k} = r^{-2k}s$$

If $a \neq 0$, we get $x \in cl(s)$ and if $a = 0$, $x \in \langle r^2 \rangle$.

Therefore, $x \in N \Rightarrow \langle r^2, s \mid sr^2s = r^{-2} \rangle \subseteq N$.

Now, let $x \in N$, $x \in cl(s)$ or $x \in \langle r^2 \rangle \subseteq \langle r^2, s \mid sr^2s = r^{-2} \rangle$. For $x \in cl(s)$, $x = r^{2k}s \in \langle r^2, s \mid sr^2s = r^{-2} \rangle$.

From above we get that $N = \langle r^2, s | sr^2s = r^{-2} \rangle$.
 In order to check that N is isomorphic to D_n (Dihedral Group of order n)
 Consider $x = r^2, y = s$, we note that order of y is 2 and ,

$$|x| = \frac{n}{\gcd(n, 2)} = \frac{n}{2}$$

Also, $xyx = (s)(r^2)(s) = r^{-2} = x^{-1}$.

Thus,

$$N = \langle x, y | x^{\frac{n}{2}} = y^2 = e, yxy = x^{-1} \rangle$$

This is precisely the Dihedral Group of order n , D_n . □

By Theorem (11), Normality of N follows.

Case 2.b: $cl(rs) \subseteq N$ Following a similar reasoning as that used for
Case 2.a, we get that $|N|$ is n .
 Clearly $r^{2k+1}s, r^{2(k+1)+1}s \in N$.

$$(r^{2(k+1)+1}s)(r^{2k+1}s) = r^2$$

Thus we again get $N = cl(rs) \cup \langle r^2 \rangle$.

Following a similar argument as that for *Case 2.a* we get

$$N = \langle r^2, rs | (rs)r^2(rs) = r^{-2} \rangle \cong D_n$$

In this case we consider, $x = r^2$ and $y = rs$. Following a similar calculation as that in the previous case yields the above result.

And normality of N follows from Theorem (11).

3.4 Quotient Groups of D_{2n}

This section contains the quotients of D_{2n} . It is easy to show any order 2 group is isomorphic to $\mathbb{Z}_2$.

3.4.1 Quotients with normal subgroup consisting of only rotations

- When $N = \langle r \rangle$. We have $|D_{2n}/N| = 2$. Hence, $D_{2n}/N \cong \mathbb{Z}_2$.
- For the case when $N = \langle r^d \rangle$ where $d \mid n$. We consider the following elements of D_{2n}/N : $R = rN$ and $S = sN$. We note,

$$(sN)^2 = (rN)^d = N = e$$

$$(sN)(rN)(sN) = (srs)N = r^{-1}N = (rN)^{-1}$$

In this case e is the identity of the quotient group. From the above results we get,

$$D_{2n}/N = \langle R, S \mid R^d = S^2 = e, SRS = R^{-1} \rangle \cong D_{2d}$$

3.4.2 Quotients with normal subgroup consisting of at least one reflection

From the previous section for an odd n we have $N = D_{2n}$. Hence, the quotient in this case is the trivial group $D_{2n}/N \cong \{e\}$. However, when n is even, the order of N is n . Therefore, $|D_{2n}/N| = \frac{2n}{n} = 2$. Hence, $|D_{2n}/N| \cong \mathbb{Z}_2$.

Chapter 4

Semi-Direct Product and Automorphism Group of D_{2n}

We shall first define the semi-direct product of two groups and then proceed to the automorphism group of D_{2n} .

4.1 Semi-Direct Product

Definition 17 (External Semi-Direct Product). *Given two groups H and K and a homomorphism $\Phi : K \rightarrow \text{Aut}(H)$. We denote the semi-direct product of H and K with respect to Φ as $H \rtimes_{\Phi} K$.*

- The underlying set is $\{(h, k) \mid h \in H, k \in K\}$.
- The group operation is $(h_1, k_1) \circ (h_2, k_2) = (h_1 \Phi(k_1)(h_2), k_1 k_2)$

We see that the identity of $H \rtimes_{\Phi} K$ is (e_H, e_K) where e_H and e_K are the identities in H and K respectively. For any $(h, k) \in H \rtimes_{\Phi} K$,

$$(h, k)(e_H, e_K) = (h \Phi(k)(e_H), (k)(e_K)) = (h, k)$$

$$(e_H, e_K)(h, k) = (e_H \Phi(e_K)(h), (e_K)(k)) = (h, k)$$

The group operation is well-defined and the inverse of any $(h, k) \in H \rtimes_{\Phi} K$ is $(\Phi(k^{-1})(h^{-1}), k^{-1})$.

$$(h, k)(\Phi(k^{-1})(h^{-1}), k^{-1}) = (h \Phi(k)(\Phi(k^{-1})(h^{-1}), k k^{-1}) = (e_H, e_K)$$

$$(\Phi(k^{-1})(h^{-1}), k^{-1})(h, k) = (\Phi(k^{-1})(h^{-1}) \Phi(k^{-1})(h), k^{-1} k) = (e_H, e_K)$$

Associativity is also satisfied. Hence, $H \rtimes_{\Phi} K$ is a group.

We can also define an internal semi-direct product if we are given two subgroups of a group where one group is a normal subgroup.

Definition 18 (Internal Semi-Direct Product). *Let H and K be two subgroups of G and a homomorphism $\Phi : K \rightarrow \text{Aut}(H)$ given by $\Phi(k)(h) = khk^{-1}$ (conjugation of h by k) where $h \in H$ and $k \in K$. We say that G is the semi-direct product of H and K written as $G = H \rtimes K$ if the following hold:*

- $G = HK$
- $H \trianglelefteq G$
- $H \cap K = \{e\}$

For an internal semi-direct product for any $g \in G$ we can uniquely write $g = hk$ where $h \in H$ and $k \in K$. While the product of any two elements say $g_1 = h_1k_1$ and $g_2 = h_2k_2$ in G is defined as $g_1g_2 = h_1(k_1h_2k_1^{-1})k_1k_2$.

Our treatment of semi-direct products may feel brisk. However, for our purpose, this much knowledge of semi-direct products will suffice and we shall not delve further into more theory about the topic.

4.2 Semi-Direct Product form of D_{2n}

We will first consider the internal semi-direct product form of D_{2n} followed by its external semi-direct form.

4.2.1 Internal Semi-Direct Product

We consider the following two subgroups of D_{2n} : $H = \langle r \rangle$ and $K = \{e, s\}$. We note that $H \cap K = \{e\}$.

From our analysis of D_{2n} in chapter 2, we know every element in D_{2n} can be written in the form $r^i s^b$ where $i \in \mathbb{Z}$ and $b \in \{0, 1\}$. Hence, $D_{2n} \subseteq HK$ and as both sets have the same cardinality we have,

$$D_{2n} = HK$$

We note that $|H| = n$ and the index of H in D_{2n} is 2. The normality of H in D_{2n} follows from Theorem (11).

$$H \trianglelefteq D_{2n}$$

By definition (18), D_{2n} is the semi-direct product of its rotational subgroup of order n and $K = \{e, s\}$.

$$D_{2n} = H \rtimes K$$

Additionally it is easy to check that conjugation map of any rotation by a reflection s is indeed an automorphism of $\langle r \rangle$.

4.2.2 External Semi-Direct Product

We now consider the following two groups: $\mathbb{Z}_n = \{0, 1, 2, \dots, n-1\}$ and $\mathbb{Z}_2 = \{0, 1\}$. We prove the following result.

Claim 8. *Consider a cyclic group of order n , $\langle a \rangle$. The map $\phi : \langle a \rangle \rightarrow \langle a \rangle$ where $\phi(a^x) = a^{-x}$ for $x \in \mathbb{Z}$ is an automorphism of $\langle a \rangle$.*

Proof. Clearly ϕ is well-defined. We first prove that ϕ is a homomorphism. Let a^x and a^y be elements of $\langle a \rangle$.

$$\phi(a^x a^y) = \phi(a^{x+y}) = a^{-(x+y)} = a^{-x} a^{-y} = \phi(a^x) \phi(a^y)$$

Hence, ϕ is a homomorphism. Also for any $a^x \in \langle a \rangle$,

$$\phi(\phi(a^x)) = \phi(a^{-x}) = a^x$$

The map ϕ is its own inverse. Therefore, ϕ is an automorphism of $\langle a \rangle$. $\square$

Note: We consider the identity of the semi-direct product $H \rtimes K$ as (e, e) where the first e is the identity in H and the e in the second component is the identity in K .

Theorem 14. *Consider the homomorphism $\Phi : \mathbb{Z}_2 \rightarrow \text{Aut}(\mathbb{Z}_n)$. We can define a semi-direct product with respect to Φ where $\Phi(1) = \phi$ where ϕ maps any element of $\mathbb{Z}_n$ to its inverse. We have the following result:*

$$D_{2n} \cong \mathbb{Z}_n \rtimes_{\Phi} \mathbb{Z}_2$$

Proof. Consider the map $\psi : \mathbb{Z}_n \rtimes_{\Phi} \mathbb{Z}_2 \rightarrow D_{2n}$ given by $\psi(h, k) = r^h s^k$ where $h \in \mathbb{Z}_n$ and $k \in \mathbb{Z}_2$.

We first prove that ψ is a homomorphism. Clearly, the identity $(0, 0)$ is mapped to identity. Let $(h_1, k_1), (h_2, k_2) \in \mathbb{Z}_n \rtimes_{\Phi} \mathbb{Z}_2$.

$$\psi((h_1, k_1)(h_2, k_2)) = \psi(h_1 + \phi(k_1)(h_2) \pmod{n}, k_1 + k_2 \pmod{2})$$

$$\psi((h_1, k_1)(h_2, k_2)) = r^{h_1 + (-1)^{k_1} h_2 \pmod{n}} s^{(k_1 + k_2) \pmod{2}}$$

And,

$$\psi(h_1, k_1) \psi(h_2, k_2) = (r^{h_1} s^{k_1})(r^{h_2} s^{k_2}) = r^{h_1 + (-1)^{k_1} h_2 \pmod{n}} s^{(k_1 + k_2) \pmod{2}}$$

Therefore, ψ is a homomorphism. We note that the cardinalities of both groups are finite and equal. Hence, it will suffice to prove injectivity.

Say $(h_1, k_1), (h_2, k_2) \in \mathbb{Z}_n \rtimes_{\Phi} \mathbb{Z}_2$ such that,

$$\psi(h_1, k_1) = \psi(h_2, k_2)$$

$$\begin{aligned} r^{h_1} s^{k_1} &= r^{h_2} s^{k_2} \\ r^{h_1-h_2} &= s^{k_2-k_1} \end{aligned}$$

We see that as no reflection is equal to a rotation and as $h_1, h_2 \in \mathbb{Z}_n$ we get $h_1 = h_2$ and $k_1 = k_2$.

$$(h_1, k_1) = (h_2, k_2)$$

Thus, ψ is an isomorphism and we get,

$$D_{2n} \cong \mathbb{Z}_n \rtimes_{\Phi} \mathbb{Z}_2$$

□

4.3 The Automorphism Group of D_{2n} , $Aut(D_{2n})$

We first prove some properties regarding automorphisms of D_{2n} .

Claim 9. *Any automorphism of D_{2n} , $\phi \in Aut(D_{2n})$ maps rotations to rotations which forces ϕ to map reflections to reflections.*

Proof. Let a rotation $r \in \langle r \rangle \leq D_{2n}$. As ϕ is an automorphism of D_{2n} it can be easily shown that ϕ preserves order of elements. Thus, $\phi(r)$ must be mapped to an element of order n . We know that every reflection has an order of 2. Therefore, $\phi(r)$ must be mapped to a rotation of order n in fact $\phi(r) = r^k$ where $\gcd(k, n) = 1$. As $\phi \in Aut(D_{2n})$,

$$\phi(\langle r \rangle) = \langle r \rangle$$

□

We now define another group called the multiplicative group of integers modulo n :

$$\mathbb{Z}_n^* = \{i \in \mathbb{Z} \mid 1 \leq i \leq n-1, \gcd(i, n) = 1\}$$

We note that the identity of this group is $1 \in \mathbb{Z}_n^*$ as $\gcd(1, n) = 1$.

Claim 10. *The map $\phi_k : \mathbb{Z}_n \rightarrow \mathbb{Z}_n$ defined as $\phi_k(x) = kx \pmod{n}$ where $k \in \mathbb{Z}_n^*$ is an automorphism of $\mathbb{Z}_n$.*

Proof. We note that the map ϕ_k is well-defined. Since the group $\mathbb{Z}_n$ is finite in order to prove bijectivity proving injectivity will suffice.

Assume $x, y \in \mathbb{Z}_n$ where,

$$\begin{aligned} \phi_k(x) &= \phi_k(y) \\ kx &\equiv ky \pmod{n} \\ x &\equiv y \pmod{n} \end{aligned}$$

The last step follows as $\gcd(k, n) = 1$. We get $x = y$ as both $0 \leq x, y \leq n$. Therefore, ϕ_k is a bijection.

We must now prove ϕ_k is a homomorphism. Let $x, y \in \mathbb{Z}_n$,

$$\phi_k(x + y) = k(x + y) \pmod{n} = kx + ky \pmod{n}$$

$$\phi_k(x) + \phi_k(y) = kx \pmod{n} + ky \pmod{n} = kx + ky \pmod{n}$$

Since, ϕ_k is bijective and a homomorphism. We conclude that ϕ_k is an automorphism $\square$

Lemma 8. *Given two groups G and H . A homomorphism $\phi : G \rightarrow H$ is injective if and only if the kernel is trivial, $\ker \phi = \{e_G\}$.*

$$\ker \phi = \{g \in G \mid \phi(g) = e_H\}$$

Proof. ($\Rightarrow$) Let ϕ be injective. For any $x \in \ker \phi$ then $\phi(x) = e_H$ but we know $\phi(e_G) = e_H$ and since ϕ is injective we have $x = e_G$.

($\Leftarrow$) Let $x, y \in G$ such that,

$$\phi(x) = \phi(y)$$

$$\phi(y)^{-1}\phi(x) = e_H$$

$$\phi(y^{-1}x) = e_H$$

As the kernel is trivial, we get $y^{-1}x = e_G \Rightarrow x = y$. Thus, ϕ is injective. $\square$

Lemma 9. *For any $(x, y) \in \mathbb{Z}_n \rtimes_{\Phi} \mathbb{Z}_n^*$ there exists an automorphism $f \in \text{Aut}(D_{2n})$ such that:*

- $f(r) = r^y$
- $f(s) = r^x s$

Proof. Using the fact that any element of D_{2n} can be written as $r^a s^b$ where $a \in \{0, 1, \dots, n-1\}$ and $b \in \{0, 1\}$. We consider the map $f : D_{2n} \rightarrow D_{2n}$ where $f(r^a s^b) = r^{b'x+ay} s^{b'}$, we take $b' \equiv b \pmod{2}$. For the sake of convenience we will write $(a+b) \pmod{2}$ as $(a+b)'$

We first prove f is a homomorphism. Let $r^{a_1} s^{b_1}, r^{a_2} s^{b_2} \in D_{2n}$.

$$f((r^{a_1} s^{b_1})(r^{a_2} s^{b_2})) = f(r^{a_1+(-1)^{b_1}a_2} s^{b_1+b_2}) = r^{(a_1+(-1)^{b_1}a_2)y+(b_1+b_2)'x} s^{(b_1+b_2)'}$$

$$f(r^{a_1} s^{b_1})f(r^{a_2} s^{b_2}) = r^{a_1y+b_1'x} s^{b_1'} r^{a_2y+b_2'x} s^{b_2'} = r^{(a_1+(-1)^{b_1}a_2)y+(b_1+(-1)^{b_1}b_2)'x} s^{(b_1+b_2)'}$$

We can easily check that f is a homomorphism for all four possible cases: $b_1 = b_2 = 0$, $b_1 = b_2 = 1$, $b_1 = 0$ and $b_2 = 1$, $b_1 = 1$ and $b_2 = 0$.

The kernel of f ,

$$\ker f = \{r^a s^b \in D_{2n} \mid f(r^a s^b) = e\}$$

$$\ker f = \{r^a s^b \in D_{2n} \mid r^{ay+b'x} s^{b'} = e\}$$

Now, $b' = 0$ as no rotation is equal to a reflection. We get, $r^{ay} = e$ therefore $n \mid ay$ but as $\gcd(y, n) = 1$ we get $n \mid a$. Now, $0 \leq a < n$ this forces $a = 0$. Hence, f has a trivial kernel and is injective. We get that f is an automorphism of D_{2n} .

It is easy to check that $f(r) = r^y$ and $f(s) = r^x s$.

□

We are now equipped to prove the following fact about the automorphism group D_{2n} .

Theorem 15. *Given the homomorphism $\Phi : \mathbb{Z}_n^* \rightarrow \text{Aut}(\mathbb{Z}_n)$ such that for $k \in \mathbb{Z}_n^*$ we have $\Phi(k) = \phi_k$ (ϕ_k defined in Claim (10)) where for any $x \in \mathbb{Z}_n$ we have $\phi_k(x) = kx \pmod{n}$.*

$$\text{Aut}(D_{2n}) \cong \mathbb{Z}_n \rtimes_{\Phi} \mathbb{Z}_n^*$$

Proof. We consider the map $\psi : \text{Aut}(D_{2n}) \rightarrow \mathbb{Z}_n \rtimes_{\Phi} \mathbb{Z}_n^*$ where if $\phi \in \text{Aut}(D_{2n})$ such that $\phi(r) = r^k$ and $\phi(s) = r^h s$ we have,

$$\psi(\phi) = (h, k)$$

We note that by the reasoning used in Claim (9), $h \in \mathbb{Z}_n$ and $k \in \mathbb{Z}_n^*$. Thus, ψ is well-defined.

We first prove that ψ is a homomorphism. Let $\phi_1, \phi_2 \in \text{Aut}(D_{2n})$ where $\phi_i(r) = r^{k_i}$ and $\phi_i(s) = r^{h_i} s$,

$$\begin{aligned} \phi_1 \circ \phi_2(r) &= \phi_1(r^{k_2}) = (\phi_1(r))^{k_2} = r^{k_1 k_2} \\ \phi_1 \circ \phi_2(s) &= \phi_1(r^{h_2} s) = (\phi_1(r))^{h_2} (\phi_1(s)) = r^{h_1 + k_1 h_2} s \\ \psi(\phi_1 \circ \phi_2) &= (h_1 + k_1 h_2, k_1 k_2) \end{aligned}$$

Writing this in more familiar notation,

$$\psi(\phi_1 \circ \phi_2) = (h_1 + \Phi(k_1)(h_2), k_1 k_2)$$

Now,

$$\psi(\phi_1)\psi(\phi_2) = (h_1, k_1)(h_2, k_2) = (h_1 + \Phi(k_1)(h_2), k_1 k_2)$$

As $\psi(\phi_1 \circ \phi_2) = \psi(\phi_1)\psi(\phi_2)$, ψ is a homomorphism.

In order to prove injectivity we shall prove that the kernel of ψ is trivial.

$$\ker \psi = \{\phi \in \text{Aut}(D_{2n}) \mid \psi(\phi) = (0, 1)\}$$

Any such ϕ satisfies $\phi(s) = s$ and $\phi(r) = r$ as r and s generate D_{2n} . We get the result $\phi(x) = x$, $\forall x \in D_{2n}$, ϕ is the identity map, id . Therefore,

$$\ker \psi = \{id\}$$

In order to prove surjectivity, for any $(x, y) \in \mathbb{Z}_n \rtimes_{\Phi} \mathbb{Z}_n^*$. By Lemma (9) there exists an automorphism, ϕ that satisfies $\phi(r) = r^y$ and $\phi(s) = r^x s$.

By the above calculations ψ is an isomorphism and we get,

$$Aut(D_{2n}) \cong \mathbb{Z}_n \rtimes_{\Phi} \mathbb{Z}_n^*$$

□

Appendix A

Proofs of Some Theorems

A.1 Proof of Normality of index-2 subgroups

Lemma 10 (Lemma). $gN = Ng, \forall g \in G$ if and only if $N \trianglelefteq G$.

Proof. ($\Rightarrow$) Consider any $x \in gNg^{-1}$. $x = gng^{-1}$, for some $n \in N$. As $gN = Ng$, we get that $gn = n'g$, where $n' \in N$.

$$x = gng^{-1} = (n'g)g^{-1} = n' \in N$$

Therefore $gNg^{-1} \subseteq N$, for all $g \in G$,

$$N \trianglelefteq G$$

($\Leftarrow$) If $N \trianglelefteq G$, then

$$gng^{-1} = n' \in N, \forall n \in N, \forall g \in G$$

Therefore, $gn = n'g \in Ng, \forall n \in N$.

$$gN \subseteq Ng$$

By normality of N in G ,

$$g^{-1}ng = n_1 \in N$$

Hence, $ng = gn_1 \in gN, \forall n \in N$.

By the above two results equality is established,

$$gN = Ng \quad \forall g \in G$$

□

Theorem 16. *Every index-2 subgroup of a group G is normal in G .*

Proof. Let H be a index-2 subgroup of G .

Then the distinct left cosets are H and gH , and the right cosets H and Hg , $g \notin H$.

As the cosets partition the group. and all the cosets have the same size.

$$G = H \cup gH = H \cup Hg$$

$$gH = G \setminus H \text{ and } Hg = G \setminus H.$$

Therefore $gH = Hg$.

Now clearly if $g \in H$, then $Hg = gH$ and if $g \notin H$ by the above derivation equality holds.

Hence we get that,

$$H \trianglelefteq G$$

□

A.2 Proof of Fundamental Theorem of Cyclic Groups

Theorem 17 (Fundamental Theorem of Cyclic Groups). *Given a cyclic group $G = \langle a \rangle$. Every subgroup $H \leq G$ is cyclic.*

Proof. Let $H \leq G$, if $H = \{e\}$ we are done. Hence, we assume that H is not the trivial subgroup.

As $H \leq G$, every element in H has the form a^i where $i \in \mathbb{Z}$. As by assumption H is not the trivial subgroup, there exists some non-zero $t \in \mathbb{Z}$ such that $a^t \in H$. As H is a subgroup $a^{-t} \in H$. Now, t or $-t$ one of them must be positive.

Therefore, there exists $a^m \in H$ where $m > 0$. By Well-Ordering Principle a least d exists such that $a^d \in H$. Clearly, $\langle a^d \rangle \leq H$ by closure property of H .

We now prove that $H = \langle a^d \rangle$.

We know every element of H has the form $a^i \in H$ where $i \in \mathbb{Z}$. If $d \mid i$ then $i = dk$ where $k \in \mathbb{Z}$.

$$a^i = a^{dk} = (a^d)^k \in \langle a^d \rangle$$

If $d \nmid i$ then $i = dq + i'$ where $q, i' \in \mathbb{Z}$ and $0 < i' < d$.

We note $a^i, (a^d)^q \in H \Rightarrow a^i(a^d)^{-q} \in H$.

$$(a^i)(a^d)^{-q} = a^{i-dq} = a^{i'} \in H$$

This contradicts the minimality of d . Therefore, every element of H is a multiple of d . We get, $H \leq \langle a^d \rangle$.

$$H = \langle a^d \rangle$$

□

Corollary 2. *Let $G = \langle a \rangle$ be a cyclic group. If G is finite where $|G| = n$ then for every divisor d of n there exists a unique cyclic group of order d . In fact the subgroups of G correspond bijectively to the divisors of n .*

Proof. For every divisor $d \mid n$, the cyclic subgroup $\langle a^{\frac{n}{d}} \rangle$ has order d .

$$|\langle a^{\frac{n}{d}} \rangle| = |a^{\frac{n}{d}}| = \frac{n}{\gcd(n, \frac{n}{d})} = d$$

Assume that another subgroup $\langle a^m \rangle$ has order d ,

$$\frac{n}{\gcd(n, m)} = d = \frac{n}{\frac{n}{d}}$$

We get $\frac{n}{d} = \gcd(n, m)$. Therefore, $\frac{n}{d} \mid m$.

$$\langle a^m \rangle \leq \langle a^{\frac{n}{d}} \rangle$$

Equality follows as both subgroups have order d .

Thus, for every divisor of n there exists a unique subgroup of order equal to that of the divisor.

Now, by Theorem (17) every subgroup of G has a form $\langle a^m \rangle$, where m is the least positive integer such that $a^m \in \langle a^{\frac{n}{d}} \rangle$.

We note that $\langle a^m \rangle = \langle a^{\gcd(n, m)} \rangle$ as $\gcd(n, m) \mid m$ we get $\langle a^m \rangle \leq \langle a^{\gcd(n, m)} \rangle$ and

$$\frac{n}{\gcd(n, m)} = \frac{n}{\gcd(n, \gcd(n, m))}$$

The order of both subgroups are equal and equality follows.

Now $\gcd(n, m)$ is clearly a divisor of n and we get that all the subgroups of G correspond bijectively to the divisors of n . $\square$

Appendix B

Generating Sets and Presentation

Given a group G and a subset $A = \{a_1, a_2, \dots, a_n\}$ of G . We define the collection of all the subgroups of G that contain A as $\mathcal{A}$.

We define a subgroup,

$$H = \bigcap_{K \in \mathcal{A}} K$$

Claim 11. H is the smallest subgroup of G that contains A .

Proof. As H is the intersection of subgroups, H is a subgroup itself.

Assume that some other subgroup J contains A then clearly $J \in \mathcal{A}$.

By the definition of H we have $H = \bigcap_{K \in \mathcal{A}} K \subset J$.

Therefore, the minimality of H follows. □

Theorem 18. $H = \langle A \rangle = \{x_1^{\epsilon_1} x_2^{\epsilon_2} \dots x_n^{\epsilon_n} \mid n \in \mathbb{N}, x_i \in A, \epsilon_i \in \{1, -1\}\}$

Proof. We note that clearly $A \subseteq \langle A \rangle$. Any $a \in A$ can be written as $a^1 \in \langle A \rangle$.

If any subgroup contains A it must contain all finite product of its elements by closure property. Hence, $\langle A \rangle \subseteq H$.

We now prove that $\langle A \rangle$ is a subgroup of G . $\langle A \rangle \neq \phi$ (if $A = \phi$ then $\langle A \rangle = \{\phi\}$)

$e = a_i(a_i)^{-1} \in \langle A \rangle$ (for any $a_i \in A$). Therefore, identity exists.

Let $a, b \in \langle A \rangle$, then $a = a_1^{x_1} a_1^{x_2} \dots a_n^{x_n}$ and $b = b_1^{y_1} b_2^{y_2} \dots b_m^{y_m}$,

$$b^{-1} = b_m^{-y_m} b_{m-1}^{-y_{m-1}} \dots b_1^{-y_1} \in \langle A \rangle$$

We see that $ab = a_1^{x_1} a_1^{x_2} \dots a_n^{x_n} b_1^{y_1} b_2^{y_2} \dots b_m^{y_m} \in \langle A \rangle$.

Thus, $\langle A \rangle$ is a subgroup that contains A and by minimality of H we have $H \subseteq \langle A \rangle$.

It follows that $H = \langle A \rangle$. □

Definition 19 (Generating Set). Given a set $A = \{a_1, a_2, \dots, a_n\}$, $n \in \mathbb{N}$. The group generated by A is defined as,

$$G = \langle A \rangle$$

We call the set A a generating set of G .

Definition 20 (generators). *Given a generating set A of a group G , we call the elements of A as the generators of the group G .*

Definition 21 (Group Presentation). *Given a generating set A of G and a set of relations, R which consists of relations among the elements of the generating set. We denote the group presentation of G by,*

$$G = \langle A \mid R \rangle$$